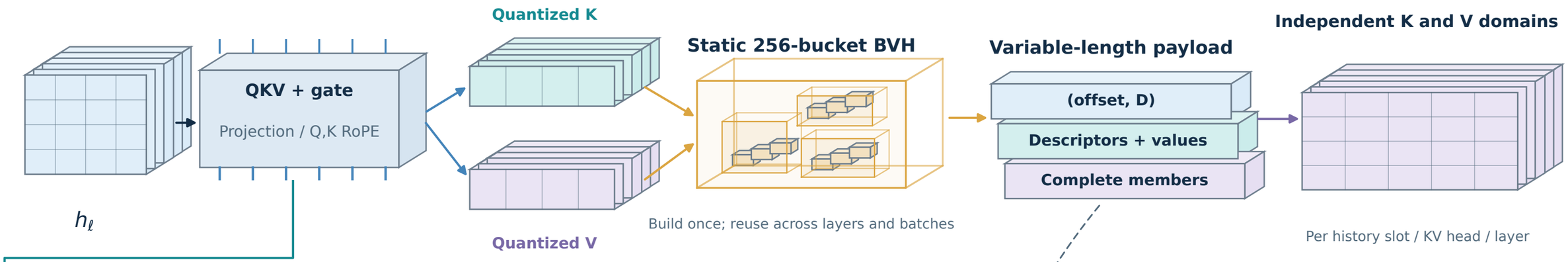
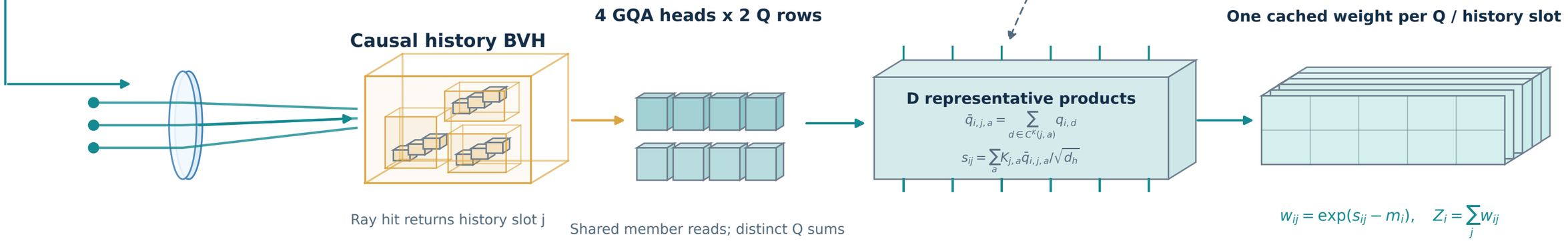
One attention layer, two hardware roles

The QKV projection precedes classification; later Q queries reuse compact class ownership

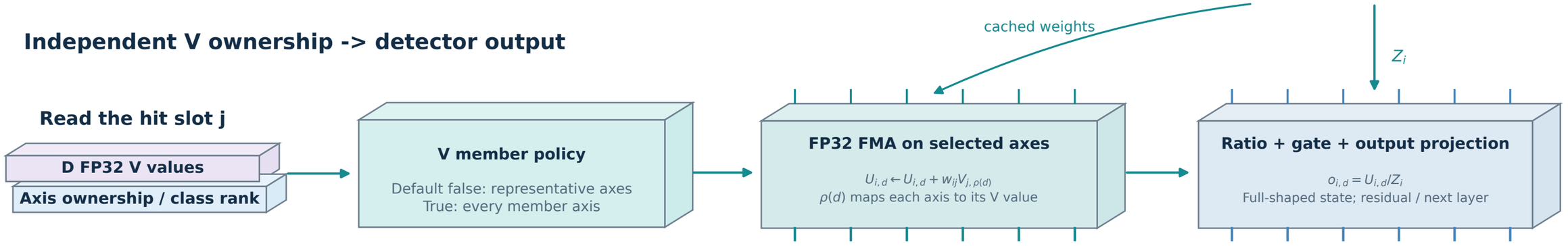
a Write K,V once; classify with a permanent scene



b RT history hit -> compact K gather -> cached weights



c Independent V ownership -> detector output



Default false selects representative axes; true evaluates every V member axis.

RT Core: BVH lookup

FP32 Shader: exact identity + score + weight + V accumulation